

mechanistic insights and model evaluations. The perceived utility of mechanistic interpretability for AI safety largely depends on researchers’ priors regarding the likelihood of these different risk scenarios.

How Could Mechanistic Insight be Harmful? Mechanistic interpretability research could accelerate AI capabilities, potentially leading to the development of powerful AI systems that are misaligned with human values, posing significant risks (Soares, 2023; Kross, 2023; Hendrycks & Mazeika, 2022). While historically, interpretability research had little impact on AI capabilities, recent exceptions like discoveries about scaling laws (Hoffmann et al., 2022), architectural improvements inspired by studying induction heads (Olsson et al., 2022; Fu et al., 2023a; Poli et al., 2023; Schuster et al., 2022), and efficiency gains inspired by the logit lens technique (Schuster et al., 2022) demonstrated its potential to enhance capabilities. Scaling interpretability research may necessitate automation (Conmy et al., 2023; Bills et al., 2023), potentially enabling rapid self-improvement of AI systems (RicG, 2023). Some researchers recommend selective publication and focusing on lower-risk areas to mitigate these risks (Hobbhahn & Chan, 2023; Shovelain & McKernon, 2023; Elhage et al., 2022b; Nanda et al., 2023a).

Mechanistic interpretability also poses dual-use risks, where the same techniques could be used for both beneficial and harmful purposes. Fine-grained editing capabilities enabled by interpretability could be used for *machine unlearning* (removing private data or dangerous knowledge from models) (Guo et al., 2024; Sun et al., 2024; Nguyen et al., 2022; Pochinkov & , 2023) but could be misused for censorship. Similarly, while interpretability may help improve adversarial robustness (Räuker et al., 2023), it may also facilitate the development of stronger adversarial attacks (Mu & Andreas, 2020; Casper et al., 2023b).

Misunderstanding or overestimating the capabilities of interpretability techniques can divert resources from critical safety areas or lead to overconfidence and misplaced trust in AI systems (Charbel-Raphaël, 2023; Casper, 2023). Robust evaluation and benchmarking (Section 8.2) are crucial to validate interpretability claims and reduce the risks of overinterpretation or misinterpretation.

7 Challenges

7.1 Research Issues

Need for Comprehensive, Multi-Pronged Approaches. Current interpretability research often focuses on individual techniques rather than combining complementary approaches. To achieve a holistic understanding of neural networks, we propose utilizing a diverse interpretability toolbox that integrates multiple methods (see also Section 4.4), such as: (i) Coordinating observational (e.g., probing, logit lens) and interventional methods (e.g., activation patching) to establish causal relationships. (ii) Combining feature-level analysis (e.g., sparse autoencoders) with circuit-level interventions (e.g., path patching) to uncover representation-mechanism interplay. (iii) Integrating intrinsic interpretability approaches with post-hoc analysis for robust understanding.

For example, coordinated methods could be used for *reverse engineering* trojaned behaviors (Casper et al., 2023c), where observational techniques identify suspicious activations, interventional methods isolate the relevant circuits, and intrinsic approaches guide the design of more robust architectures.

Cherry-Picking and Streetlight Interpretability. Another concerning pattern is the tendency to cherry-pick results, relying on a small number of convincing examples or visualizations as the basis for an argument without comprehensive evaluation (Räuker et al., 2023). This amounts to publication bias, showcasing an unrealistic highlight reel of best-case performance. Relatedly, many interpretability techniques are primarily evaluated on small toy models and tasks (Chughtai et al., 2023; Elhage et al., 2022b; Jermyn et al., 2022; Chen et al., 2023b), risking missing critical phenomena that only emerge in more realistic and diverse contexts. This focus on cherry-picked results from toy models is a form of *streetlight interpretability* (Casper, 2023), examining AI systems under only ideal conditions of maximal interpretability.

7.2 Technical Limitations

Scalability Challenges and Risks of Human Reliance. A critical hurdle is demonstrating the scalability of mechanistic interpretability to real-world AI systems across model size, task complexity, behavioral coverage, and analysis efficiency (Elhage et al., 2022b; Scherlis et al., 2023). Achieving a truly comprehensive understanding of a model’s capabilities in all contexts is daunting, and the time and compute required must scale tractably. Automating interpretability techniques is crucial, as manual analysis quickly becomes infeasible for large models. The high human involvement in current interpretability research raises concerns about the scalability and validity of human-generated model interpretations. Subjective, inconsistent human evaluations and lack of ground-truth benchmarks are known issues (Räuker et al., 2023). As models scale, it will become increasingly untenable to rely on humans to hypothesize about model mechanisms manually. More work is needed on automating the discovery of mechanistic explanations and translating model weights into human-readable computational graphs (Elhage et al., 2022b), but progress on that front may also come from outside the field (Lu et al., 2024).

Obstacles to Bottom-Up Interpretability. There are fundamental questions about the tractability of fully *reverse engineering* neural networks from the bottom up, especially as models become more complex (Hendrycks, 2023a). Models may learn internal representations and algorithms that do not cleanly map to human-understandable concepts, making them difficult to interpret even with complete transparency (McGrath et al., 2022). This gap between human and model ontologies may widen as architectures evolve, increasing opaqueness (Hendrycks et al., 2022). Conversely, model representations might naturally converge to more human-interpretable forms as capability increases (Hubinger, 2019a; Feng & Steinhardt, 2023).

Analyzing Models Embedded in Environments. Real-world AI systems embedded in rich, interactive environments exhibit two forms of in-context behavior that pose significant interpretability challenges beyond understanding models in isolation. Externally, models may dynamically adapt to and reshape their environments through in-context learning from the interactions and feedback loops with their external environment (Leahy, 2023). Internally, the *hydra effect* demonstrates in-context reorganization, where models flexibly reorganize their internal representations in a context-dependent manner to maintain capabilities even after ablating key components (McGrath et al., 2023). These two instances of in-context behavior – external adaptation to the environment and internal self-reorganization – undermine interpretability approaches that assume fixed *circuits*. For models deeply embedded in rich real-world settings, their dynamic coupling with the external world via in-context environmental learning and their internal in-context representational reorganization make strong interpretability guarantees difficult to attain through analysis of the initial model alone.

Adversarial Pressure Against Interpretability. As models become more capable through increased training and optimization, there is a risk they may learn deceptive behaviors that actively obscure or mislead the interpretability techniques meant to understand them. Models could develop adversarial "mind-reader" components that predict and counteract the specific analysis methods used to interpret their inner workings (Sharkey, 2022; Hubinger, 2022). Optimizing models through techniques like gradient descent could inadvertently make their internal representations less interpretable to external observers (Hubinger, 2019b; Fu et al., 2023b; von Oswald et al., 2023). In extreme cases, a highly advanced AI system singularly focused on preserving its core objectives may directly undermine the fundamental assumptions that enable interpretability methods in the first place.

These adversarial dynamics, where the capabilities of the AI model are pitted against efforts to interpret it, underscore the need for interpretability research to prioritize worst-case robustness rather than just average-case scenarios. Current techniques often fail even when models are not adversarially optimized. Achieving high confidence in fully understanding extremely capable AI models may require fundamental advances to make interpretability frameworks resilient against an intelligent system’s active deceptive efforts.

8 Future Directions

Given the current limitations and challenges, several key research problems emerge as critical for advancing mechanistic interpretability. These problems span four main areas: emphasizing conceptual clarity Section 8.1, establishing rigorous standards Section 8.2, improving the scalability of interpretability techniques Section 8.3, and expanding the research scope Section 8.4. Each subsection presents specific research questions and challenges that need to be addressed to move the field forward.

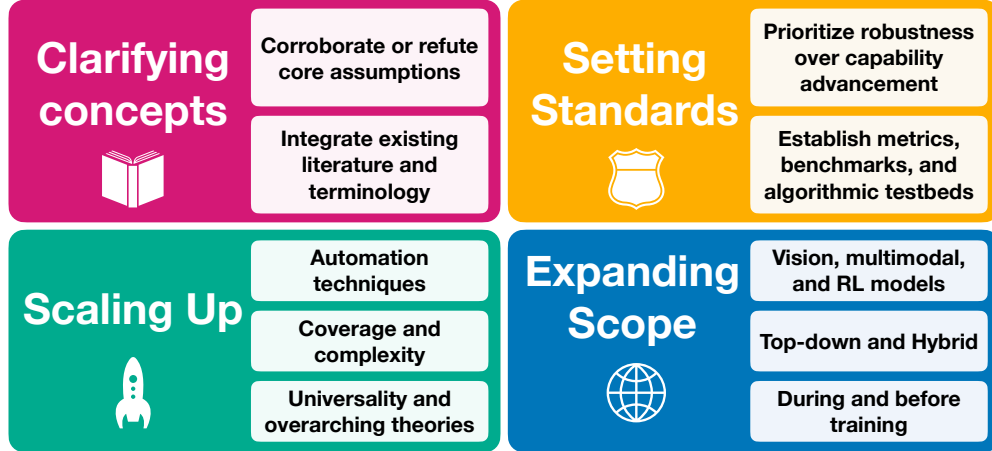


Figure 12: Roadmap for advancing mechanistic interpretability research, highlighting key strategic directions.

8.1 Clarifying Concepts

Integrating with Existing Literature. To mature, mechanistic interpretability should embrace existing work, using established terminology rather than reinventing the wheel. Diverging terminology inhibits collaboration across disciplines. Presently, the terminology used for mechanistic interpretability partially diverges from mainstream AI research (Casper, 2023). For example, while the mainstream speaks of *distributed representations* (Hinton, 1984; Olah, 2023) and the goal of *disentangled* representations (Higgins et al., 2018; Locatello et al., 2019), the mechanistic interpretability literature refers to the same phenomenon as *polysemanticity* (Scherlis et al., 2023; Lecomte et al., 2023; Marshall & Kirchner, 2024) and *superposition* (Elhage et al., 2022b; Henighan et al., 2023). Using common language invites "accidental" contributions and prevents isolating mechanistic interpretability from broader AI research.

Mechanistic interpretability relates to many other fields in AI research, including compressed sensing (Elhage et al., 2022b), modularity, adversarial robustness, continual learning, network compression (Räuker et al., 2023), neurosymbolic reasoning, trojan detection, and program synthesis (Casper, 2023; Michaud et al., 2024), and causal representation learning. These relationships can help develop new methods, metrics, benchmarks, and theoretical frameworks. For instance:

- i.) **Neurosymbolic Reasoning and Program Synthesis:** Mechanistic interpretability aims for *reverse engineering* neural networks by converting their weights into human-readable algorithms. This endeavor can draw inspiration from neurosymbolic reasoning (Riegel et al., 2020) and program synthesis. Techniques like creating programs in domain-specific languages (Verma et al., 2019b;a; Trivedi et al., 2021), extracting decision trees (Zhang et al., 2019) or symbolic causal graphs (Ren et al., 2023) from neural networks align well with the goals of mechanistic interpretability. Adopting these approaches can extend the toolkit for reverse engineering AI systems.
- ii.) **Causal Representation Learning:** Causal Representation Learning (CRL) aims to discover and disentangle underlying causal factors in data (Schölkopf et al., 2021), complementing mechanistic interpretability’s goal of understanding causal structures within neural networks. While mechanistic interpretability typically examines individual *features* and *circuits*, CRL offers a framework for